

Email Classification - Example

- A csv file containing related information of 5172 randomly picked email files and their respective labels for spam or not-spam classification.
- The csv file contains 5172 rows, each row for each email.
- There are 3002 columns.
- The first column indicates Email name.
- The last column has the labels for prediction : 1 for spam, 0 for not spam.
- The remaining 3000 columns are the 3000 most common words in all the emails, after excluding the non-alphabetical characters/words.

Email Classification - Example

A	B	C	D	E	F	G	H	I
Email No.	the	to	ect	and	for	of	a	you
Email 1	0	0	1	0	0	0	2	
Email 2	8	13	24	6	6	2	102	
Email 3	0	0	1	0	0	0	8	
Email 4	0	5	22	0	5	1	51	
Email 5	7	6	17	1	5	2	57	
Email 6	4	5	1	4	2	3	45	
Email 7	5	3	1	3	2	1	37	
Email 8	0	2	2	3	1	2	21	
Email 9	2	2	3	0	0	1	18	
Email 10	4	4	35	0	1	0	49	
Email 11	22	14	2	9	2	2	104	
Email 12	33	28	27	11	10	12	173	
Email 13	27	17	3	7	5	8	106	
Email 14	4	5	7	1	5	1	37	
Email 15	2	4	6	0	3	1	16	
Email 16	6	2	1	0	2	0	36	
Email 17	3	1	2	2	0	1	17	
Email 18	36	21	6	14	7	17	194	
Email 19	1	3	1	0	2	0	14	

emails
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Email Classification - Example

	infrastruct	military	allowing	ff	dry	Prediction
0	0	0	0	0	0	0
0	0	0	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	1	0
0	0	0	0	0	0	1
0	0	0	0	0	0	0
0	0	0	0	0	1	1
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	1	0
0	0	0	0	0	5	0
0	0	0	0	0	4	0
0	0	0	0	0	2	0
0	0	0	0	0	1	0
0	0	0	0	0	0	0
0	0	0	0	0	1	1
0	0	0	0	0	3	1
0	0	0	0	0	0	0

Email Classification - Example

- For each row, the count of each word(column) in that email(row) is stored in the respective cells.
- Thus, information regarding all 5172 emails are stored in a compact data frame rather than as separate text files.
- What are the parameters to calculate/estimate?

Email Classification - Example

When the n input attributes X_i each take on J possible discrete values, and Y is a discrete variable taking on K possible values, then our learning task is to estimate two sets of parameters.

The first is:

$$\theta_{ijk} \equiv P(X_i = x_{ij} | Y = y_k)$$

for each input attribute X_i , each of its possible values x_{ij} , and each of the possible values y_k of Y .

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In addition, we must estimate parameters that define the prior probability over Y :

$$\pi_k \equiv P(Y = y_k)$$

We can estimate these parameters using either maximum likelihood estimates (based on calculating the relative frequencies of the different events in the data), or using Bayesian MAP estimates (augmenting this observed data with prior distributions over the values of these parameters)

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Maximum likelihood estimates given a set of training examples D are given by

$$\hat{\theta}_{ijk} = \hat{P}(X_i = x_{ij} | Y = y_k) = \frac{\#D\{X_i = x_{ij} \wedge Y = y_k\}}{\#D\{Y = y_k\}}$$

where the $\#D\{x\}$ operator returns the number of elements in the set D that satisfy property x .

In the program define a matrix:

e.g.:

```
probability_matrix[i][j][k]
```

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- One danger of this maximum likelihood estimate is that it can sometimes result in MLE estimates of zero, if the data does not happen to contain any training examples satisfying the condition in the numerator.
- To avoid this, it is common to use a “smoothed” estimate which effectively adds in a number of additional “hallucinated” examples, and which assumes these hallucinated examples are spread evenly over the possible values of X_i . This smoothed estimate is given by

$$\hat{\theta}_{ijk} = \hat{P}(X_i = x_{ij} | Y = y_k) = \frac{\#D\{X_i = x_{ij} \wedge Y = y_k\} + l}{\#D\{Y = y_k\} + lJ}$$

- where J is the number of distinct values X_i can take on, and l determines the strength of this smoothing (i.e., the number of hallucinated examples is lJ).

Email Classification - Example

- Maximum likelihood estimates:

$$\hat{\pi}_k = \hat{P}(Y = y_k) = \frac{\#D\{Y = y_k\}}{|D|}$$

- where $|D|$ denotes the number of elements in the training set D .
- Example:

```
float prob_class[2]
```

Implementation Hint:
Use log likelihood and smoothing.